Ep 5 · Document AI Pipeline — Cheat Sheet

AI System Design · AI Engineering with Ajit · github.com/byruajit/ai-system-design

PIPELINE

1. Upload → object store → **queue** (return a job ID)
2. **Split pages** → classify each page
3. **Extract per doc type** → value · confidence · page
4. **Schema** validation (types, formats)
5. **Rules + cross-checks** (income vs payslip, LTV, FOIR)
6. Route: auto-accept or **field-level human review**
7. **Idempotent** write to the system of record + provenance
8. Corrections → **eval dataset**

THREE LAYERS OF TRUST

Schema → business rules → confidence. **Confidence alone is not enough.**

SIZING

- pages/s = pages/day ÷ window seconds
- workers = pages/s × seconds per page (+ 2–3× for peaks)
- Northwind: 2.8 pages/s → ~6 workers

OCR + LLM VS VISION LLM

Use	Pick
Typed / digital PDFs	OCR + small LLM
Photos, stamps, handwriting	Vision LLM
Decide by	field accuracy on 200 real pages

METRICS

Field precision/recall · **error rate in auto-accepted fields** · straight-through rate · review time per file

RELIABILITY

Idempotency key = hash(file) + pipeline version · DLQ for poison files · reprocess by version, never overwrite reviewed values

PII

Mask Aadhaar (last 4) · least pages to the model · logs carry field names, not values

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Northwind Bank is fictional. Prices are illustrative tiers — check current provider pricing.